

Question	Marking Guidance	Mark	Comments
07.1	DNA v tRNA 1. Deoxyribose v ribose; 2. Double-stranded v single-stranded; 3. Many nucleotides v few ; 4. Thymine v uracil; 5. Linear v clover leaf (structure) OR <u>Double</u> helix v clover (leaf structure); 6. Does not bind to amino acid v does bind to amino acid; 7. No exposed bases v <u>anticodon</u> ;	3 max (3 x AO1)	Apply list rule 2. Accept double helix for double stranded 3 Accept longer v shorter 4. Ignore T and U 5. Accept description of clover leaf structure. 5. Accept straight for linear 6. Accept attach/carry/have binding site for 'bind to'

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07.2	1. Use centrifuge/centrifugation at slow/low/increasing (sequence of) speed(s); 2. Large/dense organelles (removed) in (first/early) pellet OR Less dense organelles (removed) in supernatant OR Small organelles (removed) in supernatant;	2 (1 x AO2, 1 x AO3)	Ignore homogenate OR filtering Ignore cold, isotonic, buffered 2. Accept descriptions of supernatant and pellet, eg liquid and sediment/bottom of tube 2. Accept light for small 2. Accept ribosomes for small organelles 2. Accept more dense OR heavy for large 2 Accept named large organelle in pellet, eg mitochondria

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07.3	(Tube A) 1. (Ribosomes bound to) rough endoplasmic reticulum; 2. (Are) denser/heavier so move further; (Tube B) 3. (Only free ribosomes because) membrane/phospholipids/endoplasmic reticulum dissolved (by detergent);	3 (3 x AO2)	Award marks across A and B 1. and 3 Do not credit ER/rER the first time used if endoplasmic reticulum not given 2. Accept, free ribosomes are lighter so form a band higher in tube OR do not move as far 2. For 'heavier/lighter' accept 'denser/less dense'. 2. Accept settle lower OR are lower down OR at bottom OR in pellet, for move further 3. Accept broken down for dissolved 3. Ignore damaged for dissolved

